

US 20230287600A1

(54) METHOD OF ENZYMATIC PROCESSING OF PLANT BIOMASS TO PRODUCE TEXTILE GRADE FIBER	(30) Foreign Application Priority Data Jul. 2, 2020 (IN) 20201028274
(71) Applicant: GENCREST PRIVATE LIMITED, Mumbai (IN)	Publication Classification
(72) Inventors: Rashmi CHOWDHARY, Mumbai (IN); Anshika AGARWAL, Mumbai (IN); Nitin SHETYE, Mumbai (IN); Deepak KULKARNI, Mumbai (IN)	(51) Int. Cl. D01C 1/02 (2006.01)
	(52) U.S. Cl. CPC D01C 1/02 (2013.01)
(21) Appl. No.: 18/013,911	(57) ABSTRACT The current invention discloses an enzyme-based method for making high quality textile fibres from plant derived bio-mass. The invention discloses a method for production of high-quality textile grade fibres that have no loss in quality parameters as compared to textile grade fibres made from cellulosic biomass by conventional methods that use harsh chemical treatments. The fibres produced by this method from raw natural fibres can be spun into yarn by automated procedures and the yarn can be woven into high quality fabrics by powerloom as well as handloom.
(22) PCT Filed: Jul. 2, 2021	
(86) PCT No.: PCT/IB2021/055948 § 371 (c)(1), (2) Date: Dec. 30, 2022	

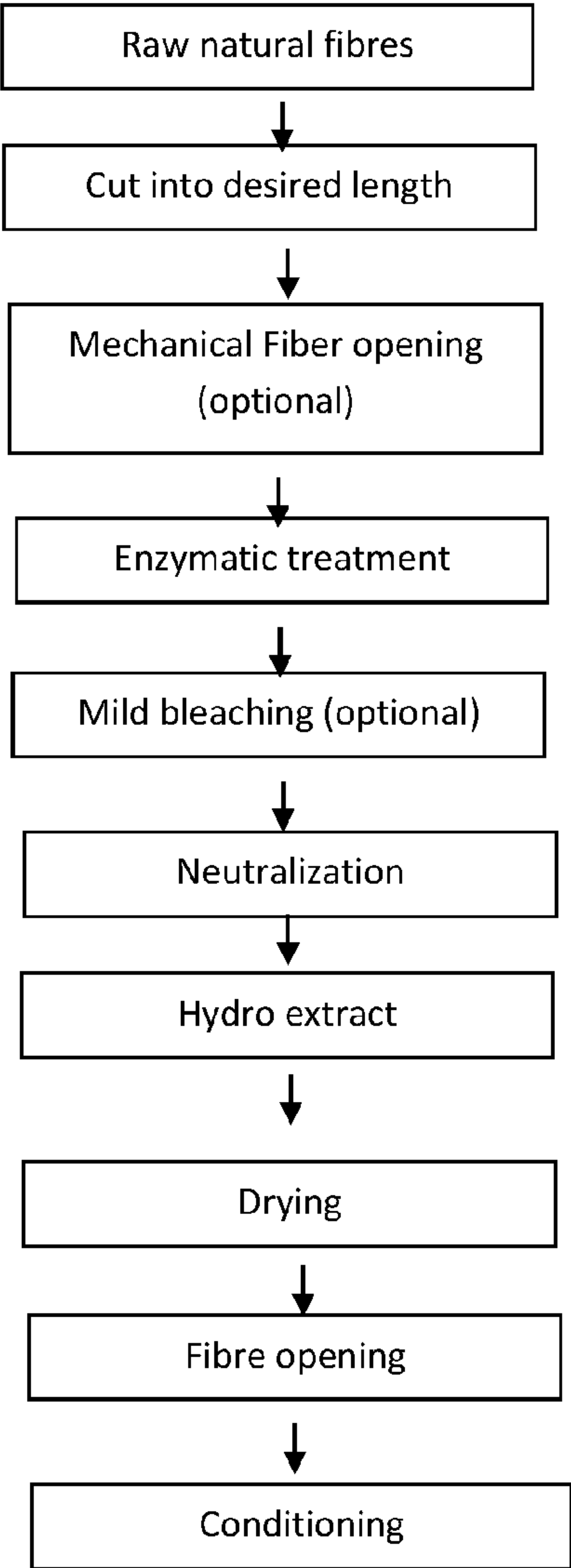
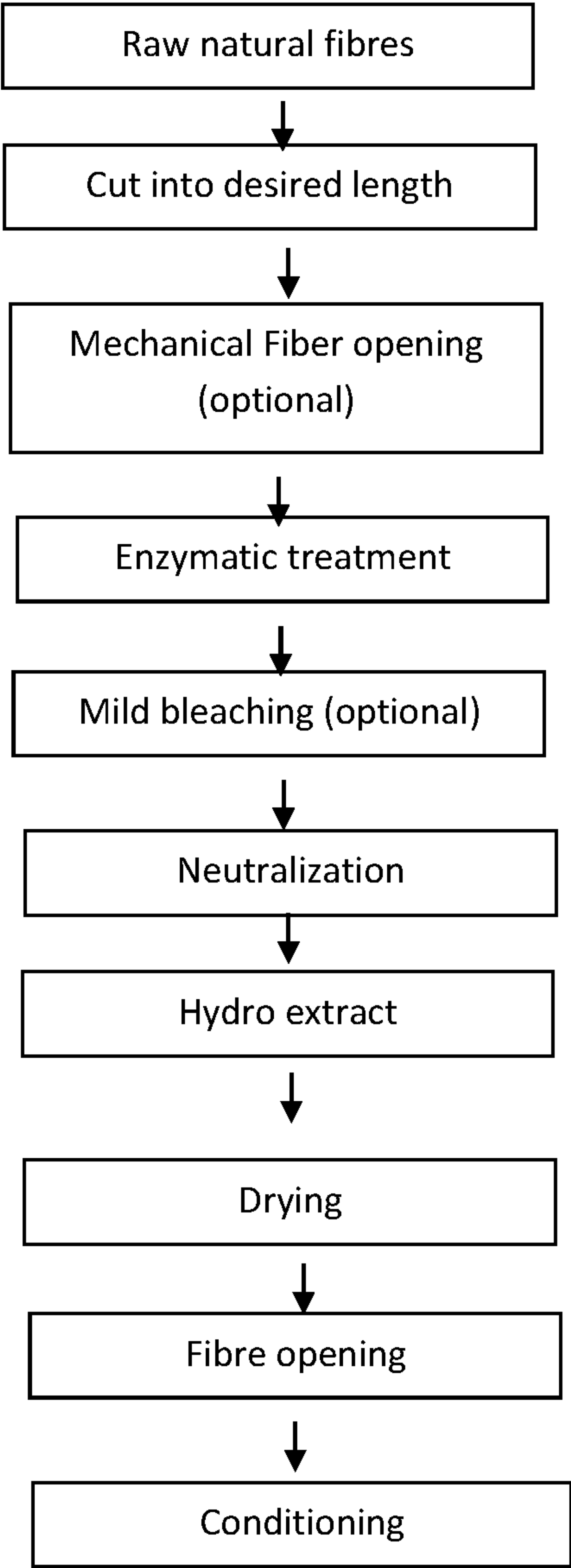


Fig 1



METHOD OF ENZYMATIC PROCESSING OF PLANT BIOMASS TO PRODUCE TEXTILE GRADE FIBER

FIELD OF INVENTION

[0001] The current invention relates to the field of obtaining textile grade fibres from plant derived biomass using an enzymatic method, and specifically a method for processing various types of raw natural fibres to obtain textile grade fibres by enzymatic means. The current invention particularly relates to making high quality textile grade fibre by processing cellulosic biomass by using a method with enzymatic activities, without using harsh chemical or treatment.

BACKGROUND

[0002] The demand for textile fibres is increasing constantly due to increasing world population, and the demand for fibres has almost doubled in the past 16 years or so. Along with providing food and clothing for the increasing world population, there is increasing need for conserving land and resources. Thus, efforts are being made to get fibres and food products from the same crops land resources. Moreover, synthetic petroleum-based fibres are non-biodegradable, hence present a different challenge to the ecosystem. These challenges can be met if high quality, textile grade fibres can be produced from various types of plant derived biomass, a lot of which is also a by-product of the agricultural practices. Cellulose, which is the most abundantly available organic matter on earth, is, in its natural and regenerated form a major source of fibre for textile industry. Plants are the major source of natural cellulosic fibres, these cellulosic fibres can be used for production of different types of fibres for various industrial applications. These cellulosic fibres may be from plants grown primarily for the fibres or from plants in which the fibres are primarily considered a by-product, for example coconut, sugarcane, banana, and pineapple. Globally, farming generates millions of tons of agricultural by-products each year. Some of the by-products are used as animal feed and for other small-scale applications. Many of the agricultural by-products contain substantial amounts of cellulose, especially in fibrous form. The by-product-type fibres have not been used extensively for several reasons including limited availability, difficulty in extraction, lesser performance in terms of spinning properties, lower quality fibres that cannot be used for making high end products, and limited growing regions. Utilizing agricultural by-products to a larger extent for downstream applications can provide benefit growers economically and provide environmental benefits by reducing the amount of by-product disposal. Moreover, the products made using these agricultural by-products are 100% bio-degradable with added qualities such as—breathability, hypoallergenicity, and high hygroscopicity.

[0003] Since many of the agricultural by-products contain substantial amounts of cellulose, especially in fibrous form, they can be used to produce fibres that can be used in various industries. But natural fibres which are extracted from plant biomass are generally coarser and have high lignin and hemicellulose content. This makes these fibres very rigid and thick and hence limits its application to very few sectors such as handicrafts, handlooms, upholstery. The process of conversion of such thick and rigid fibres into more softer fibres involves chemical treatments which are very harsh

and pose environmental concerns. The quality of the fibres achieved so far by conventional methods using just enzymes is not sufficient for textile application due to difficulty in spinning it into yarn, and requires further harsh chemical steps to make it into spinnable grade, but with other drawbacks such as brittleness and loss in luster of the fibres.

[0004] Although enzymes are known to have several advantages over chemical or mechanical processes for textile fibre production, attempts to make textile grade fibres from raw natural fibres by using an entirely enzymatic process has been unsuccessful till now. Most of the processes using enzymes use harsh chemical treatments also in the process, or use high amounts of enzymes for long durations. Moreover, these processes have not been shown to produce high quality fibres, from raw fibres, which can be spun into yarn by automatic methods, and the resultant yarn can be woven by using powerlooms too. The material made from these processes are many times crude and are hand-woven. References like CN107385523A (7) and CN108977895A (8) describe use of enzymes to produce fibres from hemp and corn, but these processes include pretreatment with acid and alkali, and have longer treatments with high concentrations of enzymes, and yet do not yield fibres with as fine denier, and spinnable into yarns by power looms, as does the method disclosed in the current invention.

[0005] The current invention presents an eco-friendly method of enzymatically converting plant derived biomass into high quality textile grade fibres, without using harsh chemical or mechanical treatments. The method disclosed herein mainly uses enzymes, and produces high quality textile fibres that are usually obtained in conventional methods by harsh chemical and/or mechanical treatments.

SUMMARY

[0006] The current invention encompasses a method of converting raw natural fibres to textile grade fibres, the method comprising the step of enzymatically treating raw natural fibres obtained from plant derived biomass to convert them to textile grade fibres, wherein the fibre denier of the textile grade fibre is not more than 75 dpf (denier per fibre) and breaking strength is at least 50 g. In one embodiment, the breaking strength of the fibre is 175 g.

[0007] In one embodiment, the plant derived biomass comprises of plants selected from group consisting of banana, pineapple, hemp, nettle, flex, jute, sisal, and remi.

[0008] In one embodiment, the total enzyme concentration in the fibre treatment container is 0.5-1%.

[0009] In one embodiment, the enzymatic treatment of the raw natural fibres to convert them to textile grade fibres is done for 2 to 4 hours.

[0010] In one embodiment, the enzymes used for the step of enzymatic treatment of raw natural fibres are selected from the group consisting of cellulase, pectinase, hemicellulase, lignase, esterase, amylase, and peroxidase.

[0011] In one embodiment, the method further comprises alkaline protease treatment of the raw natural fibres before the step of enzymatic treatment.

[0012] In one embodiment, the alkaline protease concentration present in the fibre treatment container is 0.1-0.3%.

[0013] In one embodiment, the method further comprises the step of mechanically or manually opening the raw natural fibres before the enzymatic treatment.

[0014] In one embodiment, the plant derived biomass is not mechanically or manually treated to open the raw natural fibres before the enzymatic treatment.

[0015] In one embodiment, the method does not comprise any chemical treatment step.

[0016] In one embodiment, the method does not comprise any chemical pre-treatment step before the enzymatic step.

[0017] In one embodiment, the method further comprises the step of mild chemical treatment after the enzymatic treatment.

[0018] In one embodiment, the textile grade fibres produced from method disclosed herein are spinnable into yarn by automated methods. In one embodiment, the yarn produced is woven or knitted into fabrics. In one embodiment, the yarn is woven into fabric using power loom or handloom.

[0019] In one embodiment, the yarn made from the textile grade fibres produced from method disclosed herein is woven or knitted.

[0020] In one embodiment, the fibres are made into fabrics without blending with other types of fibres.

[0021] In one embodiment, the textile grade fibres made by the method disclosed herein, are blended with other man-made or natural fibres to create yarn of at least 20 count.

[0022] In one embodiment, the current invention encompasses textile grade fibres produced from method disclosed herein, wherein they have high tensile strength, good elongation, high tenacity, and finer diameter

BRIEF DESCRIPTION OF DRAWINGS

[0023] FIG. 1 shows the flowchart showing the steps in the method of producing textile grade fibres from plant derived biomass disclosed in the current invention.

DETAILED DESCRIPTION

[0024] The current invention discloses method for enzymatic production of textile grade fibres from plant derived biomass. The current invention particularly relates to making high quality textile grade fibre by processing cellulosic fibre by an enzyme-based method, without using harsh chemical treatments. This method is eco-friendly, and moreover, the fibres produced are as high quality as produced by conventional means from cotton.

[0025] Cellulose is the most abundantly available organic matter on earth, is, in its natural and regenerated form a major source of fibre for textile industry. Plants are the major source of natural cellulosic fibres. These cellulosic fibres can be used for production of different types of fibres for various industrial applications. Globally, farming generates millions of tons of agricultural by-products each year. Many of the agricultural by-products contain substantial amounts of cellulose, especially in fibrous form. Utilizing agricultural by-products to a larger extent can provide benefit growers economically and provide environmental benefits by reducing the amount of by-product disposal. Moreover, the products made using these agricultural by-products are 100% bio-degradable. The by-product-type fibres have not been used extensively for several reasons including limited availability, difficulty in extraction, low quality fibre as end product which limits use for high end products, performance-related problems, such as low spinnability, coarse and hard texture.

[0026] Fibres from the stalks or stems and leaves of plants are conventionally extracted by a process known as “retting.” A traditional retting process is dew retting, which utilizes bacteria and fungi in the environment to delignify the fibres to convert them into suitable state for use by conventional textile machinery. Dew-Retting can yield inconsistent results, gives poor fibre quality, and usually can only be performed in limited geographical regions, and occupies agricultural land during the retting process. Chemical methods for retting have been used and these typically produce fibres that are more consistent and with improved physical properties, but these chemical methods raise many environmental concerns including waste disposal concerns. Apart from this, finished fibre products are of low quality in terms of brightness index and low tensile strength.

[0027] The current invention provides an enzymatic method for converting plant based or plant derived biomass, into high grade textile fibres. The method disclosed herein uses low quantities of enzyme, and does not involve alkali or acid pretreatment of raw natural fibres. The method is short duration of enzymatic treatment, and yields spinnable fibres of very high-quality indices. The fibres can be spun into yarn by automated methods, and the resultant yarn can be woven into fabrics by powerloom also, hence can be used to make fabrics/textiles at industrial scale.

Definitions

[0028] The term “plant derived biomass” as used herein is defined as biomass extracted from plants. It can be from plants that are specifically grown for obtaining that biomass, or can be a by product of the main crop. Plant-derived biomass can be from any plant, including naturally growing plants, or agricultural crops.

[0029] The term “raw natural fibre” as used herein is defined as are the fibres which are mechanically extracted from plant biomass using manual processor machine extractors. Examples of raw natural fibres includes, without limitation, fibres derived from Banana, hemp, Bamboo, nettle, flex, pineapple, remi, jute, kenaf, sesal, abacca, coconut but not limited to these.

[0030] Raw fibres are derived from plant biomass by machine extractors, the machine strips the biomass open and extract the bast fibres.

[0031] The term “degumming” as used herein is defined as the process of separating fibres from each other by removing biomolecules such as pectins, starch, hemicellulose, gums and other biomolecules which binds the fibre together.

[0032] The terms “fibre treatment container” and “treatment bath” and “bath” are used interchangeably herein and refer to the container or the bath in which the enzymatic treatment of the raw fibres is done. The percentage of the enzyme used and the amount of raw fibres added to the bath would depend on the volume of the bath. The liquid inside the container is also referred to herein as “liquor” or “liquid”. It might be water or any other aqueous solvent with additives for optimal action of enzymes.

[0033] The major constituent of the plant cell wall is “lignocelluloses”, which consists of lignin (15-20%), hemicellulose (25-30%) and cellulose (40-50%). These components together form a three-dimensional complex network bound by covalent and non-covalent interactions.

[0034] Lignocellulose is generally found, for example, in the fibres, pulp, stems, leaves, hulls, canes, husks, and/or cobs of plants or fibres, leaves, branches, bark, and/or wood

of trees and/or bushes. Examples of lignocellulosic materials include, but are not limited to, agricultural biomass, such as farming and/or forestry material and/or residues, branches, bushes, canes, forests, grains, grasses, short rotation woody crops, herbaceous crops, and/or leaves; crop residues, such as corn, millet, and/or soybeans, herbaceous material and/or crops; forests; fruits; flowers; needles; logs; roots; saplings; shrubs; switch grasses; vegetables; fruit peels; vines; wheat midlings; oat hulls; hard and soft woods; or any combination thereof.

[0035] Hemicelluloses consist of xylan, a heteropolysaccharide substituted with monosaccharides such as L-arabinose, D-galactose, D-mannoses and organic acids such as acetic acid, ferulic acid, glucuronic acid interwoven together with help of glycosidic and ester bonds. Depolymerization of this complex polymer is essential for its efficient utilization in different industrial application.

[0036] The term “raw natural fibre” as used herein is defined as are the fibres which are mechanically extracted from plant biomass using manual process or machine extractors. Examples of raw natural fibres includes, without limitation, fibres derived from Banana, hemp, Bamboo, nettle, flex, ramie, jute, kenaf, sesal, abacca, coconut but not limited to these.

[0037] Raw fibres are derived from plant biomass by machine extractors, the machine strips the biomass open and extract the bast fibres.

[0038] The terms “enzymatic formulation”, “enzyme cocktail” and “enzymatic composition” are used interchangeably herein, and refer to a formulation comprising at least one enzyme, and it does not comprise any chemicals.

[0039] “Cellulase” or “cellulases”, as used herein, refer to an enzyme capable of hydrolyzing cellulose to glucose. Non-limiting examples of cellulases include mannan endo-1,4- β -mannosidase, 1,3- β -D-glucan glucanohydrolase, 1,3- β -glucan glucanohydrolase, 1,3-1,4- β -D-glucan glucanohydrolase and 1,6- β -D-glucan glucanohydrolase.

[0040] “Xylanase” or “xylanases”, as used herein, refer to an enzyme capable of hydrolysing xylan to xylobiose and xylotriose.

[0041] Xylanase is a group of enzymes consisting of endo-1,4- β -d-xylanases (EC 3.2.1.8), β -d-xylosidases (E.C. 3.2.1.37), α -glucuronidase (EC 3.2.1.139) acetyl xylan esterase (EC 3.1.1.72), α -l-arabinofuranosidases (E.C. 3.2.1.55), p-coumaric esterase (3.1.1.B10) and ferulic acid esterase (EC 3.1.1.73) involved in the depolymerization of xylan into simple monosaccharide and xylooligosaccharides.

[0042] The term “pectinase” includes any acid pectinase enzyme. Pectinases are a group of enzymes that hydrolyse glycosidic linkages of pectic substances mainly poly-1,4-alpha-D-galacturonide and its derivatives which enzyme is understood to include a mature protein or a precursor form thereof, or a functional fragment thereof, which essentially has the activity of the full-length enzyme. Furthermore, the term pectinase enzyme is intended to include homologues or analogues of such enzymes. Pectinases can be classified according to their preferential substrate, highly methyl-esterified pectin or low methyl-esterified pectin and polygalacturonic acid (pectate), and their reaction mechanism, beta-elimination or hydrolysis. Pectinases can be mainly endo-acting, cutting the polymer at random sites within the chain to give a mixture of oligomers, or they may be exo-acting, attacking from one end of the polymer and

producing monomers or dimers. Several pectinase activities acting on the smooth regions of pectin are included in the classification of enzymes provided by the Enzyme Nomenclature (1992) such as pectate lyase (EC 4.2.2.2), pectin lyase (EC 4.2.2.10), polygalacturonase (EC 3.2.1.15), exo-polygalacturonase (EC 3.2.1.67), exo-polygalacturonate lyase (EC 4.2.2.9) and exo-poly-alpha-galacturonosidase (EC 3.2.1.82).

[0043] Laccases, belong to the enzyme family of multicopper oxidases (MCOs), are classified as benzenediol oxygen reductases (EC 1.10.3.2) and are also known as urushiol oxidases and p-diphenol oxidases. They are considered versatile enzymes capable of oxidizing a large number of phenolic and non-phenolic molecules due to their low substrate specificity, using oxygen as electron acceptor and generating water as a by-product.

[0044] Endo-1,4- β -d-mannanase (EC 3.2.1.78) catalyzes the random cleavage of β -d-1,4-mannopyranosyl linkages within the main chain of galactomannan, glucomannan, galactoglucomannan, and mannan. They liberate short-chain β -1,4-manno-oligomers, which can be further hydrolyzed to mannose by β -mannosidases (EC 3.2.1.25).

[0045] α Amylases are starch hydrolases. Amylases are responsible for hydrolysis of starch to oligosaccharides. α -Amylase hydrolyzes the 1,4- α -glucoside bonds in compounds involving three or more molecules of glucose. β -Amylase liberates (mainly) β -maltose from starch and other compounds.

[0046] In one embodiment, the amylases used in the current invention are alpha-amylases.

[0047] As used herein, enzyme activity is defined in units. 1 unit of enzyme (U) is the amount of enzyme that catalyzes the reaction of 1 μ mol of substrate per minute.

[0048] Activity definitions of enzymes used in the current invention are given below. The substrates used for activity assay can be any substrates known for the given enzymes. Moreover, the enzymes used can be from any of the known sources.

[0049] Cellulase:

[0050] One unit of activity corresponded to the quantity of enzyme releasing 1 μ mol of dextrose reducing sugar (in glucose equivalents) per minute per ml under the assay conditions. Any suitable substrate can be used for assessing the activity. The substrate used herein is Carboxymethylcellulose sodium salt-low viscosity (SIGMA-ALDRICH).

[0051] Xylanase:

[0052] One unit of activity corresponds to the quantity of enzyme releasing 1 μ mol of xylose reducing sugar (in glucose equivalents) per minute per ml under the assay conditions. The substrate used herein for assessing xylanase activity is Xylan from beechwood (SRL).

[0053] Pectinase:

[0054] One unit of activity corresponds to the quantity of enzyme releasing 1 μ mol of D-galacturonic acid reducing sugar (in glucose equivalents) per minute per ml under the assay conditions. The substrate used herein is Pectin from apple (SIGMA).

[0055] Polygalacturonase:

[0056] One unit of activity corresponded to the quantity of enzyme releasing 1 μ mol of D-galacturonic acid reducing sugar (in glucose equivalents) per minute per ml under the assay conditions. The substrate used for assessing the activity in the current invention is Polygalacturonic acid sodium salt (SIGMA).

[0057] Amylase:

[0058] One unit of activity corresponded to the quantity of enzyme releasing 1 umol of maltose reducing sugar (in glucose equivalents) per minute per ml under the assay conditions. The substrate used herein for assessing the activity in the current invention is Potato starch soluble (HIMEDIA).

[0059] Mannanase:

[0060] One unit of activity corresponded to the quantity of enzyme releasing 1 umol of mannose reducing sugar (in glucose equivalents) per minute per ml under the assay conditions. The substrate used for assessing the activity in the current invention is Locust bean gum from *Ceratonia siliqua* seeds (SIGMA).

[0061] Laccase:

[0062] 1 unit is defined as amount of enzyme required to oxidize 1 micromole of guaiacol per min. The substrate used for assessing the activity in the current invention where the substrate is Guaiacol (SIGMA-ALDRICH).

[0063] The terms “textile grade fibre” and “textile fibre” are used interchangeably herein, and refer to fibres that are soft and have characteristics that are suitable for making textiles.

[0064] Tensile strength and breaking elongation are two of the most important mechanical properties for a textile grade fibre. Fibre tensile strength is often expressed by tenacity with a unit of force per denier or tex.

[0065] The term “cellulosic fibre” as used herein is defined as fibres comprising at least 20% of cellulose, and are made with ethers or esters of cellulose, which are of plant origin and can be obtained from the bark, wood or leaves of plants, or from other plant parts. In addition to cellulose, the fibres may also contain other components such as pectins, hemicellulose, lignin as majorly apart from other minor constituents. With different sources percentages of these components vary altering the mechanical properties of the fibres.

[0066] The term “man-made fibres” as used herein is defined as fibres that do not exist in nature, and are usually made from various chemicals, or are regenerated from plant fibres. Examples of man-made fibres, include polyester; polyamide—(nylon); acrylics; viscose, regenerated cellulosic fibres such as rayon, bamboo fibre, Lyocell, Modal, diacetate fibre, triacetate fibre, but are not limited to these.

[0067] The term “Denier”, as used herein, is used to describe the fineness of a textile material that is quantified as the materials weight in grams per 9,000 meters of that material.

[0068] The denier values given herein are denier values for each fibre or filament. Theoretically, Denier can be measured for the yarn also, but the denier values given herein refer to denier values for each textile grade filament (fibre) produced by the method disclosed herein. is used on two- and single-filament fibres. Some common calculations are as follows:

$$\begin{aligned} 1 \text{ denier} &= 1 \text{ g/9000 m} \\ &= 0.11 \text{ mg/m} \end{aligned}$$

[0069] In practice, measuring 9000 meters is both time-consuming and unrealistic. Generally, a sample of 900 meters is weighed, and the result is multiplied by ten to obtain the denier weight (9).

[0070] The term elongation is defined as the final length increase as a percentage of the starting length. The elastic elongation is of decisive importance since textile products without elasticity would hardly be useable. They must be able to deform and also return to shape. Higher elongation of textile fibres leads to easier processing and also increase comfort during wear.

[0071] The term “degree of reflectance” of a fibre, as used herein, is defined as the measure of the fraction of light that is reflected by a material or its reflectance.

[0072] The term “breaking strength” of a fibre, as used herein, is defined as the maximum amount of tensile stress that the material can withstand before breaking or deformation.

[0073] Tensile strength is a measure of maximum force attained in breaking a fibre. Tenacity is a measure the same force in relation to the linear density of the fibre or yarn. The ratio of load required to break the specimen and the linear density of that specimen is called tenacity.

[0074] Breaking strength is the force applied to break the fibres to cross-sectional area is known as breaking strength.

[0075] A loom is a device for weaving threads for getting cloth. A loom produces fabric by interlacing a series of lengthwise, parallel yarns with a series of width wise parallel yarns.

[0076] Handloom: A hand loom is a simple machine, powered by hand, and used for weaving.

[0077] A power loom is a type of loom that is powered mechanically instead of using human power to weave patterns or thread into cloth.

Embodiments

[0078] The current invention encompasses a method of converting raw natural fibres to textile grade fibres, the method comprising the step of enzymatically treating raw natural fibres obtained from plant derived biomass to convert them to textile grade fibres, wherein the fibre denier of the textile grade fibre is not more than 75 dpf (denier per fibre) and breaking strength is at least 50 g. In one embodiment, the breaking strength of the fibre is 175 g.

[0079] In one embodiment, the plant derived biomass comprises of plants selected from group consisting of banana, hemp, sisal, pineapple, nettle, flex, jute, and remi.

[0080] In one embodiment, the total enzyme concentration in the fibre treatment container is 0.5-1%.

[0081] In one embodiment, the enzymatic treatment of the raw natural fibres to convert them to textile grade fibres is done for 2 to 4 hours.

[0082] In one embodiment, the enzymes used for the step of enzymatic treatment of raw natural fibres are selected from the group consisting of cellulase, pectinase, hemicellulase, lignase, esterase, amylase, or peroxidase.

[0083] In one embodiment, the method further comprises alkaline protease treatment of the raw natural fibres before the step of enzymatic treatment.

[0084] In one embodiment, the alkaline protease is present in the fibre treatment container at the concentration of 0.1-0.3%.

[0085] In one embodiment, the method further comprises the step of mechanically or manually opening the raw natural fibres before the enzymatic treatment.

[0086] In one embodiment, the plant derived biomass is not mechanically or manually treated to open the raw natural fibres before the enzymatic treatment.

[0087] In one embodiment, the method does not comprise any chemical treatment step.

[0088] In one embodiment, the method does not comprise any chemical pre-treatment step.

[0089] In one embodiment, the method further comprises the step of mild chemical treatment after the enzymatic treatment.

[0090] In one embodiment, the textile grade fibres produced from method disclosed herein are spinnable into yarn by automated methods. In one embodiment, the yarn produced is woven or knitted into fabrics. In one embodiment, the yarn is woven into fabric using power loom or hand-loom.

[0091] In one embodiment, the yarn made from the textile grade fibres produced from method disclosed herein is woven or knitted.

[0092] In one embodiment, the fibres are made into fabrics without blending with other types of fibres.

[0093] In one embodiment, the textile grade fibres made by the method disclosed herein, are blended with other fibres selected from cellulosic fibres, linen, cotton, tencel or many more to create yarn of at least 20 count.

[0094] In one embodiment, the current invention encompasses textile grade fibres produced from method disclosed herein, wherein they have high tensile strength, good elongation, high tenacity, and finer diameter than fibres produced other conventional methods of processing raw natural fibres from agro-waste.

[0095] In one embodiment, the textile grade banana fibres are blended at a ratio of 1:3 with other fibres.

[0096] In one embodiment, the method disclosed above further comprises the step of mechanical treatment of raw natural fibres before the enzymatic treatment. In one embodiment, the method disclosed above further comprises the step of mild chemical treatment of raw natural fibres after the enzymatic treatment. In one embodiment, the mild chemical treatment is done by bleaching followed by neutralization of the bleaching agent.

[0097] In one embodiment, the chemical treatment is followed by removing excess water from the fibres or hydro extract. In one embodiment, the method further comprises the step of conditioning the fibres using softeners after hydro extract. In one embodiment, the softeners are Silicon softeners, or Amphoteric softeners (e.g. Siligen, Sapamine, Zylon SFC). In one embodiment, the softeners used are amphoteric softeners.

[0098] One embodiment is the method of converting raw natural fibres to textile grade fibres, the method comprising the steps of: (a) opening up the fibre bundle from the plant biomass into smaller fibre bundles by mechanical process (b) subjecting the small fibre bundles to at least one enzymatic step, the enzymes are selected from the group consisting of pectinase, protease, cellulase, hemicellulase, amylase, and mannanase to produce textile grade fibres from the small raw fibre bundles; and (c) softener treatment of the fibres to produce textile grade fibres, wherein the fibre diameter is not more than 75 denier, and fibre breaking strength is not less than 175 g.

[0099] In one embodiment, the method comprises the step of performing enzymatic degumming of the fibres, wherein the biomolecules such as pectins, hemi cellulose, lignin are removed.

[0100] In one embodiment, examples of plant parts that can be used for extracting the plant derived biomass include, but are not limited to, stems, leaves, and pseudostems.

[0101] In one embodiment stems from banana, ramie, bamboo are used as plant derived biomass to extract raw plant fibres.

[0102] In one embodiment, the raw natural fibres are cellulose based fibres, wherein they comprise at least 20% of cellulose. In one embodiment, the raw natural fibres also comprise pectins, hemicellulose, lignin.

[0103] In one embodiment, the plant derived biomass may be agricultural waste products such as banana pseudostem, cornhusk, cornstalk, rice straw, sorghum leaves, sorghum stems, soybean straw, wheat straw, cotton stems, and barley straw, and combinations thereof.

[0104] In one embodiment, the method disclosed herein is used for converting banana, hemp or jute fibres into textile grade fibres.

[0105] In one embodiment, raw fibres from plant derived biomass or plants can be used directly also for processing by the method disclosed herein, without any machine extraction into smaller fibre bundles.

[0106] In one embodiment, the enzymes used for the step of enzymatic treatment of raw natural fibres are selected from the group consisting of cellulases, pectinases, hemicellulases.

[0107] In one embodiment, the enzymatic activities to convert raw natural fibres into textile grade fibres are pectinolytic activity, hydrolytic activity, and oxido-reductase activity by the method disclosed herein.

[0108] In one embodiment, hydrolytic enzymatic activity removes non-cellulosic components from the plant biomass.

[0109] In one embodiment, the oxido-reductase activity improves brightness of fibres.

[0110] In one embodiment, the method disclosed herein further comprises the step of mild chemical treatment after the enzymatic treatment for enhancing whiteness index.

[0111] In one embodiment, the method disclosed herein further comprises the step of mild chemical treatment after the enzymatic treatment for enhancing whiteness index, and the mild chemical treatment is done by using alkali and/or peroxide.

[0112] One embodiment of the current invention encompasses the textile grade fibres produced by the method given herein.

[0113] In one embodiment, the textile grade fibres produced by the method disclosed herein have high tensile strength, good elongation, better spinnability, high tenacity, low resistance, finer diameter and high brightness index compared to the fibres obtained by chemical processing of raw natural fibres.

[0114] In one embodiment, the fibre diameter of the textile grade fibre produced by the method disclosed herein, is not more than 75 denier.

[0115] In one embodiment, the breaking strength of the textile grade fibre produced from hemp is not less than 50 g.

[0116] In one embodiment, the breaking strength of the textile grade fibre produced from banana is not less than 175 g.

[0117] In one embodiment, the tenacity or the strength of the fibre produced by the method described herein is not less than 2.5 grams/denier.

[0118] In one embodiment, the elongation of the fibre produced by the method described herein is not less than 5%.

[0119] In one embodiment, the brightness (degree of reflectance) is not less than 65.

[0120] In one embodiment, the textile grade fibres produced by the method disclosed herein are used for producing any material including, but not limited to, apparel, suiting—shirting, upholstery, handicrafts, doll hairs, and footwear.

[0121] In one embodiment, the fibres produced by the method disclosed herein are spun into a yarn.

[0122] In one embodiment, powerloom is used for making fabrics from the yarn spun from the textile grade fibres produced by the method disclosed herein.

[0123] In one embodiment, handloom is used for making fabrics from the yarn spun from the textile grade fibres produced by the method disclosed herein.

[0124] In one embodiment, the yarn from the textile grade fibres produced by the method disclosed herein, are further woven or knitted into fabric.

[0125] In one embodiment, the fibres produced by the method disclosed herein is used to make non-woven products.

[0126] In one embodiment, the textile grade fibres produced by the method disclosed herein are further blended with other natural or man-made fibres, examples of which include, but are not limited to, regenerated cellulosic fibres, linen, cotton, and man made fibres.

[0127] In one embodiment, the method of making textile grade fibres from plant derived biomass comprises at least

blended at the ratio of 1:1 with other fibres to produce fabric.

In one embodiment, the yarn from the fibres produced from banana by the method disclosed herein are blended at the ratio of 1:3 with other fibres to produce fabric.

[0130] In one embodiment, the total time taken for the enzymatic treatment is 2-4 hours.

[0131] In one embodiment, no detergent is used for treating the raw fibres for processing them enzymatically into textile grade fibres.

[0132] In one embodiment, no retting step by chemical or water treatment is done in the enzymatic method disclosed herein.

[0133] In one embodiment, no chelating agent is used for any of the treatment steps in the enzymatic method disclosed herein.

[0134] In one embodiment, no acid treatment step is done in the enzymatic method disclosed herein to process raw natural fibres into textile grade fibres.

[0135] In one embodiment, no alkali treatment step is done in the enzymatic method disclosed herein to process raw natural fibres into textile grade fibres.

[0136] In one embodiment, an optional alkali treatment step is done after the enzymatic treatment to bleach fibres produced by the enzymatic process to convert raw natural fibres into textile grade fibres.

[0137] Table 1 highlights the characteristics of the method disclosed herein.

TABLE 1

Fibre produced by the current method	Alkali pre-treatment			High temp. treatment (above 70° C.)	Enzyme conc. used	Chelating agent	Detergent	Time of enzymatic treatment	MLR ratio for fibre treatment
	Retting	pre-treatment	Acid pre-Treatment						
Banana fibres	No	No	No	No	0.5-1% of water bath	No	No	2-4 hours	1:10 to 1:15
Hemp fibres	No	No	No	No	1-2%	No	No	2-4 hours	1:10 to 1:15

one step of enzymatic processing In one embodiment, textile grade fibres are produced from banana pseudostems using the method disclosed herein. In one embodiment, the method of making textile grade fibres from banana biomass comprises one step of enzymatic processing banana fibres made by the method disclosed herein. In one embodiment, the step comprises treatment with enzymes selected from the group consisting of hemicellulase, pectinase, endocellulase, lignin oxidase and Mannanase.

[0128] In one embodiment, textile grade fibres are produced from hemp biomass using the method disclosed herein. In one embodiment, the method of making textile grade fibres from hemp biomass comprises two steps of enzymatic processing. banana fibres made by the method disclosed herein. In one embodiment, the first step comprise treatment with a protease. In one embodiment, the second treatment is done with enzymes selected from the group consisting of hemicellulase, pectinase, endocellulase, lignin oxidase and Mannanase.

[0129] In one embodiment, the yarn from the fibres produced from hemp by the method disclosed herein are

EXAMPLES

[0138] The enzyme activities were assayed as given in the detailed description (references 1-6). The sources of the enzymes used for the current experiments were:

[0139] Cellulase/xylanase: *Trichoderma reesei*

[0140] Pectinase, polygalaturonase and lipase: *Aspergillus niger*

[0141] Mannnanase: *Bacillus* sp.

[0142] Laccase: fungal source with min activity of 25 U/ml.

Example 1

[0143] A) Banana Fibre:

[0144] Material: Raw banana fibre from banana pseudostem (100-120 denier in diameter)

[0145] Composition: Banana fibres are made of cellulose (40-45%), hemicellulose (12-15%), lignin (10-12%), Pectin & waxes (2-3%) and other substances (30-35%).

[0146] Process Conditions:

[0147] Raw banana fibres obtained were cut to approximately 40 mm length. Fibers were then loaded in the stainless steel tube of launderometer (Model no. WFT8X500, JK & PC Texlab) for processing in a step wise manner as explained in FIG. 1, with process conditions and time duration explained in Table 2.

[0148] 1st step, Enzyme: Enzymes used for enzymatic processing were hemicellulase (500000 IU/Kg of fibre), pectinase (100000 IU/Kg of fibre), endocellulase (50000 IU/Kg of fibre), lignin oxidase (10000 IU/Kg of fibre) and Mannanase (20000 IU/Kg of fibre). Enzyme composition containing these enzymes was added as a dose of per 5 gram per liter to fibre dipped in water. The treatment was given for 2 hours at optimum condition of 50° C. and pH 5.

[0149] After the treatment the water was drained and refilled.

[0150] 2nd step, Bleaching: Bleaching was performed with solution of 0.5% alkali NaOH and H₂O₂ at 100° C. for 1 hour.

[0151] 3rd step, Hot wash: After bleaching bath was drained and refilled and hot wash was given at 85° C. temp for 10 minutes. After treatment, the bath was drained.

[0152] 4th step, Neutralization treatment was done with acetic acid at 50° C. temp for 10 minutes.

[0153] After this the water was drained.

[0154] 5th Step: Softener treatment was given with softener at pH 5.0 and temperature 50° C. for 30 minutes. Drained the bath after treatment.

[0155] After processing is completed, fibres were squeezed and finally air dried (upto moisture content 8-10% was attained).

[0156] 6th Step After drying, fibres were opened manually to separate fibres stuck to each other.

TABLE 2

Condition for processing step for banana fibre treatment			
STEPS	PROCESS	CONDITIONS	DURATION
Step 1	Enzymatic	MLR: 1:15, pH 4-5, 50° C.	2 Hr
Step 2	Bleaching	MLR: 1:15, 100° C.	1 Hr
Step 3	Hot wash	MLR: 1:15, 85° C.	10 min
Step 4	Neutralization	MLR: 1:15, 50° C.	10 min
Step 5	Softener	pH 5.0, 50° C.	30 min

[0157] Results:

[0158] Banana Fibers

TABLE 3

Sample	Fineness (Fiber Denier)	Single fibre strength (ASTM 322-01/ ISO 5079-1995)				
		ASTM* D1577-01/IS	Breaking strength	Breaking extension	Calculated Tenacity	CV (%)**
Description	234-1977	(g)	(%)	(g/d)	Strength	Extension
1 Raw banana fibres	113.2	530.37	11.77	4.68	36.3	22.7
2 Treated fibres	57.3	193.30	7.36	3.37	55.9	39.0

*American Society for Testing and Materials (ASTM), is an international standards organization

**The Coefficient of Variation (CV) is the standard Deviation (SD) expressed as a percentage of the mean CV
% = (SD ÷ mean) × 100. It is the same as elongation measurement for the fibres.

[0159] Textile grade banana fibres made with enzymatic method followed by bleaching treatment showed excellent fibre properties in terms of fineness, softness and whiteness by visual assessment as well as in test results shows excellent fibre fineness and elongation without losing fibre strength.

Example 2

[0160] Material: Raw hemp fibre (140-160 denier)

[0161] Composition of raw fibres from hemp: cellulose (70-75%), hemicellulose (15-20%), lignin (3.5-5.7%), Pectin (0.8%), water soluble matter (1.6%) and other substances (such as fats & wax, 1.2-6.2%).

[0162] Process Conditions:

[0163] Raw fibres obtained from stem of hemp plants were cut to 40 mm length. Fibers were then loaded (150 ml of liquor for 10 g of fibre) in the stainless steel tube of launderometer for processing in a step wise manner as explained in FIG. 1, with process conditions and time duration explained in Table 4.

[0164] 1st step, Enzyme treatment 1: The first step of enzymatic treatment was an alkaline protease (50000 U/Kg of fibre). The fibres were dipped in water and dosing of enzyme given was 2 gpl. The treatment was given for 1 hour at 60° C. and pH 10. After treatment fibres were neutralized with acetic acid at 50° C. for 10 minutes.

[0165] 2nd step, Enzyme treatment 2: Enzymes used for enzymatic processing step 2 is a cocktail of hemicellulase (500000 IU/kg of fibre), pectinase (75000 IU/Kg of fibre), endocellulase (20000 IU/kg of fibre), lignin oxidase (2000 IU/Kg of fibre) and lipase (5000 IU/kg of fibre). Enzyme cocktail was added as per 5 gpl dosing to the fibre dipped in water. Optimum condition maintained for enzyme cocktail processing step is 50° C. and pH 5-6. Treatment was given for 2 hrs. Water was drained and refilled.

[0166] 3rd step, Bleaching: Bleaching is performed with 0.5% solution of alkali NaOH and dosing of H₂O₂. Process is carried out in the same processing chamber for 1 hour.

[0167] After treatment wash fibres with water at 85° C. for 10 minutes. Drain the bath and refilled with water

[0168] 4th step, Neutralization: Further the bath was neutralized with acetic acid at 50° C. temp for 10 minutes.

[0169] 5th Step softener treatment was given with softener at pH 5.0 and temperature 50° C. for 30 minutes. Drain the bath after treatment.

[0170] After processing is completed, fibres were squeezed and finally air dried (upto moisture 8-10%).

[0171] 6th Step After drying fibres are opened manually to separate fibres stuck to each other

TABLE 4

Condition for processing step for hemp fibre treatment			
STEPS	PROCESS	CONDITIONS	DURATIONS
Step 1	Enzymatic 1	MLR: 1:15, pH 10.0, 60° C.	1 Hr
Step 2	Enzymatic 2	MLR: 1:15, pH 5-6, 50° C.	2 Hr
Step 3	Bleaching	MLR: 1:15, 100° C.	1 Hr
Step 4	Neutralization	MLR: 1:15, 50° C.	10 mins
Step 5	Softener	pH 5.0, 50° C.	30 mins

[0172] Results:

TABLE 5

Sample	Fineness (Fibre Denier)	Single fibre strength (ASTM 3822-07(2014))				
	ASTM D1577-2007/IS	Breaking strength	Breaking extension	Calculated Tenacity	CV (%)	
	Description	234-2003	(g)	(%)	(g/d)	Strength Extension
1	Raw hemp fibres	156.70	Test was not possible for this type of raw fibres			
3	Treated hemp fibres	43.84	57.12	7.12	1.30	44.60 38.96

CONCLUSION

[0173] Hemp fibres treated by the method disclosed herein showed good results in terms of fineness, softness and fibre clarity by visual assessment. Test result shows good result for fibre fineness and breaking strength. Fibre length also not reduced after treatment.

REFERENCES

[0174] 1. T. K Ghosh, measurement of cellulase activities, Pure & Appl. Chem., Vol. 59, No. 2, pp. 257-268, 1987

[0175] 2. Bailey, M. J., Biely, P. and Poutanen, K. (1992) Interlaboratory testing of methods for assay of xylanase activity. J. Biotechnol. 23, 257-270.

[0176] 3. Biz A, Farias F C, Motter F A, de Paula D H, Richard P, Krieger N, et al. (2014) Pectinase Activity Determination: An Early Deceleration in the Release of Reducing Sugars Throws a Spanner in the Works! PLoS ONE 9(10): e109529. <https://doi.org/10.1371/journal.pone.0109529>

[0177] 4. Alexander V. Gusakov, Elena G. Kondratyeva, Arkady P. Sinitsyn, “Comparison of Two Methods for Assaying Reducing Sugars in the Determination of Carbohydrase Activities”, International Journal of Analytical Chemistry, vol. 2011, Article ID 283658, 4 pages, 2011. <https://doi.org/10.1155/2011/283658>.

[0178] 5. Miller G. L. Use of Dinitrosalicylic acid reagent for determination of reducing sugars. Analytical Chemistry 31(3) 426-428 1959

[0179] 6. Rehan A. Abd El Monssef, Enas A. Hassan, Elshahat M. Ramadan, “Production of laccase enzyme for

their potential application to decolorize fungal pigments on aging paper and parchment, Annals of Agricultural Sciences, Volume 61, Issue 1, Pages 145-154, 2016,

[0180] 7. Patent application number CN107385523A and

[0181] 8. Patent application number CN108977895A

[0182] 9. https://www.apparelsearch.com/definitions/miscellaneous/denier_measurement_definition.htm

We claim:

1. A method of converting raw natural fibres to textile grade fibres, the method comprising the step of enzymatically treating raw natural fibres obtained from plant derived biomass to convert them to textile grade fibres, wherein the

textile grade fibre is not more than 75 denier per filament and breaking strength is at least 50 g.

2. The method of claim 1, wherein the plant derived biomass comprises of plants selected from group consisting of banana, hemp, nettle, flex, jute, and remi.

3. The method of claim 1, wherein the total enzyme concentration in fibre treatment container is 0.5-1%.

4. The method of claim 1, wherein the enzymatic treatment is done for 2 to 4 hours.

5. The method of claim 1, wherein the enzymes used for the step of enzymatic treatment of raw natural fibres are selected from the group consisting of cellulase, pectinase, hemicellulase, lignase, esterase, amylase, and peroxidase.

6. The method of claim 1, wherein it comprises alkaline protease treatment of the raw natural fibres before the step of enzymatic treatment.

7. The method of claim 6, wherein the alkaline protease is present in the fibre treatment container at the concentration of 0.1-0.3%.

8. The method of claim 1, wherein the method does not comprise any chemical treatment step.

9. The method of claim 1, wherein it further comprises the step of bleaching after the enzymatic treatment.

10. The textile grade fibres produced from method of claim 1, wherein the fibres are spinnable into yarn by automated process.

11. The textile grade fibres of claim 10, wherein the yarn produced from the fibres is woven or knitted.

12. The textile grade fibres from claim 10, wherein the yarn is woven into fabrics by powerloom or handloom.

13. The textile grade fibres derived from method of claim **1**, wherein they are blended with other man-made or natural fibres and spun to create yarn of at least 20 count.

* * * * *